

PURE MATHEMATICS WITH MECHANICS • UPPER SIXTH

UPPER SIXTH — PURE MATHEMATICS WITH MECHANICS

CHAPTER 17

INTRODUCTION TO MOTION OF CONNECTED PARTICLES

SUGGESTED TIME: 9 PERIODS

IN THIS CHAPTER

This chapter introduces the dynamics of two or more bodies linked by strings, rods or contact, moving together under gravity, friction and applied forces. Students learn to model connected systems (smooth and rough surfaces, inclined planes, pulleys and Atwood's machine), to compute accelerations and tensions, and to solve real-life problems using both Newton's laws and energy methods.

- By the end of this chapter I can model a system of connected particles with clear assumptions (light inextensible string, smooth pulley, particle model).
- By the end of this chapter I can draw separate force diagrams for each particle of a connected system.
- By the end of this chapter I can apply Newton's second law to each particle and solve the simultaneous equations for acceleration and tension.
- By the end of this chapter I can analyse motion of connected particles on smooth horizontal surfaces.
- By the end of this chapter I can include limiting and non-limiting friction ($F = \mu R$) in connected-particle problems on rough surfaces.
- By the end of this chapter I can resolve forces on an inclined plane and analyse particles connected over a pulley on slopes.



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CONTENTS

- | | |
|--|---|
| 1. Section 1: Modelling Connected Particles and Tension in Strings | eration |
| 2. Section 2: Motion on Smooth and Rough Horizontal Surfaces | 6. Section 6: Energy Methods and Real-Life Problems |
| 3. Section 3: Motion on Inclined Planes | 7. Integration activity |
| 4. Section 4: Atwood's Machine and Multi-Stage Motion | 8. Methods and worked examples |
| 5. Section 5: Pulley Systems with More Than Two Masses and Variable Accel- | 9. Exercises |
| | 10. Solutions to the exercises |
| | 11. Summary sheet |

CHAPTER OBJECTIVES

- By the end of this chapter I can model a system of connected particles with clear assumptions (light inextensible string, smooth pulley, particle model).
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- By the end of this chapter I can include limiting and non-limiting friction ($F = \mu R$) in connected-particle problems on rough surfaces.
- By the end of this chapter I can resolve forces on an inclined plane and analyse particles connected over a pulley on slopes.
- By the end of this chapter I can solve Atwood's machine problems, including the force on the pulley support.
- By the end of this chapter I can handle systems with more than two masses and multi-stage motion (string going slack, particle hitting the ground).
- By the end of this chapter I can use the work-energy principle and conservation of mechanical energy as an alternative method for connected systems.

- By the end of this chapter I can interpret and solve real-life connected-particle problems in the style of GCE Paper 2 questions.

PREREQUISITES

- Newton's three laws of motion and $F = ma$ (Lower Sixth mechanics)
- Resolving forces into components and equilibrium of a particle
- Friction: coefficient of friction, $F \leq \mu R$, limiting equilibrium
- Kinematics with constant acceleration: the suvat equations
- Work, kinetic energy, potential energy and the work-energy principle

1 SECTION 1: MODELLING CONNECTED PARTICLES AND TENSION IN STRINGS

At the Mile 4 motor park in Limbe, a mechanic uses a rope passing over a beam to lift a heavy engine out of a lorry: the engine goes down slowly while a counterweight of scrap metal rises. Why does the engine not crash to the ground, and how heavy must the counterweight be? On the Bamenda escarpment, a pickup tows a broken-down taxi up a steep slope with a chain — will the chain hold? These everyday situations all involve **connected particles**, and by the end of this chapter you will be able to predict their motion, the forces in the connectors, and the energy involved, exactly as required at the GCE Advanced Level.

In this first section we build the language of the whole chapter: what a connected-particle system is, the modelling assumptions that make it tractable, the nature of **tension**, and a five-step routine that turns any such problem into a pair of simultaneous equations.

1.1 What is a Connected-Particle System?

So far in mechanics you have mostly studied a single particle moving under given forces. In real life, bodies rarely move alone: a locomotive pulls wagons, a crane cable lifts a container, two climbers are tied together by a rope. Whenever two or more bodies are linked by a string, rope, chain, rod or tow-bar, the motion of one body constrains the motion of the others: they cannot move independently.

DEFINITION Connected particles

Two or more bodies are called **connected particles** when they are linked by a connector (string, rope, chain, rod or tow-bar) in such a way that the motion of one body determines, wholly or partly, the motion of the other(s). In the standard model, all particles attached to the same taut, inextensible string move with the same speed and the same magnitude of acceleration.

Think of the mechanic's rope in Limbe: if the engine descends by 30 cm, the counterweight must rise by exactly 30 cm, because the rope does not stretch. The two motions are locked together — that is the whole point of the model.

1.2 Modelling Assumptions

Real ropes have mass, stretch a little, and rub on rough pulleys. To obtain clean mathematics we idealise. Each assumption below has a precise physical consequence that you must be able to state in an examination.

KEY POINTS The standard modelling assumptions

- Particle model: each body is treated as a point mass. Its size, shape and rotation are

ignored; all forces act at a single point.

- Light string (massless): the string has no mass, so no net force is needed to accelerate the string itself. Consequence: the **tension** has the same value at every point of the string. (Proof: for a massless string, $F_{\text{net}} = ma = 0$, so the forces at its ends are equal and opposite.)
- Inextensible string: the string does not stretch. Consequence: the distances moved, the speeds and the magnitudes of the accelerations of all particles attached to the same taut string are equal.
- Smooth pulley or peg: there is no friction in the pulley (and the pulley is light). Consequence: the tension is the same on both sides of the pulley; the pulley only changes the direction of the force.

WARNING Inextensibility is a kinematic condition

A common error is to treat the two particles as independent and give them different accelerations. If the string is taut and inextensible, this is impossible: the string would have to stretch or break. Always write one single unknown a for the common magnitude of acceleration, and one single unknown v for the common speed.

1.3 Tension in Strings

Take a taut string and pull it at both ends. Every small piece of the string is pulled by its neighbours; this internal pulling force, transmitted along the string, is the **tension**.

DEFINITION Tension

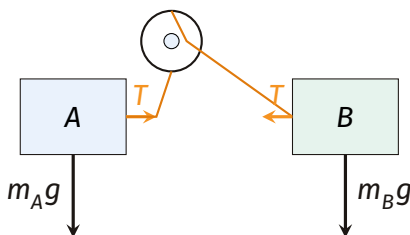
The **tension** T in a taut string is the pulling force that the string exerts on each body attached to it, directed along the string, away from the body (a string can only pull). If the string is light and passes over a smooth pulley, then T has the same value throughout the string.

Why does a string always pull and never push? Because a flexible string cannot sustain compression: push it and it simply buckles and goes slack. When a string goes slack, $T = 0$ and the particles cease to be connected — this is a classic examination trap.

PROPERTY Newton's third law and tension

The forces exerted by a taut string on the two bodies at its ends form a Newton third-law

pair: they are equal in magnitude and opposite in direction. If the string pulls body A with force T towards the string, it pulls body B with force T towards the string at the other end. (Proof not examinable; it follows directly from the light-string assumption and $F = ma$ applied to the massless string.)



Free-body diagrams of two particles connected by a light inextensible string over a smooth pulley. The tension T has the same magnitude on both sides.

Strings versus rods and tow-bars.

A string, rope or chain can only pull: the force in it is always a tension ($T \geq 0$), and it may go slack. A rigid **rod** or **tow-bar** can also push: the force in it can be a tension (pulling) or a **thrust** (pushing, sometimes written $T < 0$ or stated separately as a thrust). When a car tows a trailer with a rigid tow-bar and then brakes hard, the bar may be in thrust, pushing the trailer back. Always read the question to see which connector is used.

1.4 The Five-Step Routine

Every connected-particle problem in this chapter is solved by the same disciplined routine.

METHOD Solving a connected-particle problem

1. State the assumptions: particles, light inextensible string (or light rigid rod), smooth pulley (or smooth/rough surface, as given).
2. Draw a separate free-body diagram for each particle, showing only the forces acting on that particle: weight, normal reaction, tension(s), applied forces, friction.
3. Choose the direction of motion for each particle (consistent with the string) and mark the common acceleration a .
4. Write Newton's second law $F = ma$ for each particle, resolved along its own direction of motion. (Alternatively, write the equation for the whole system to find a directly, then use one particle's equation to find T .)
5. Solve the simultaneous equations for a and T (and check the sign of T : for a string, $T \geq 0$).

0; if $T < 0$, the string is slack and the model breaks down).

WARNING One diagram per particle!

The most frequent mistake at GCE is to draw both particles on a single diagram and to mix internal forces (the tensions) with external forces (weights, driving forces, resistances). For the whole system the internal tensions cancel out, but for each particle separately the tension is essential. Always draw two distinct diagrams and write two distinct equations.

1.5 First Example: Two Particles Over a Smooth Fixed Pulley

Consider two particles of masses m_1 and m_2 ($m_1 > m_2$) tied to the ends of a light inextensible string passing over a smooth fixed pulley — the mechanic's situation in Limbe, idealised. The heavier mass m_1 descends, the lighter mass m_2 rises, both with acceleration of magnitude a . Applying the routine:

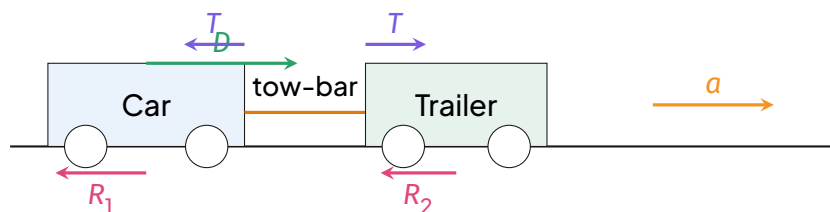
- Particle 1 (moving down, take down as positive): $m_1g - T = m_1a$.
- Particle 2 (moving up, take up as positive): $T - m_2g = m_2a$.

Adding eliminates T :

$$a = \frac{(m_1 - m_2)g}{m_1 + m_2}, \quad T = \frac{2m_1m_2g}{m_1 + m_2}.$$

Note that $a < g$: this is why the engine does not crash to the ground — the counterweight, through the tension, holds it back. The full quantitative study of this Atwood machine comes in Section 4.

1.6 Worked Example: A Car Towing a Trailer



Car towing a trailer: driving force D and resistance R_1 on the car, resistance R_2 on the trailer, and the tow-bar force T pulling the trailer forward and the car backward.

WORKED EXAMPLE 1 Car towing a trailer on a horizontal road

A car of mass 1200 kg tows a trailer of mass 400 kg along a straight horizontal road by means of a rigid tow-bar. The car's engine produces a driving force of 2400 N. Resistance forces of 600 N act on the car and 200 N on the trailer. Find (a) the acceleration of the system, (b) the force in the tow-bar, stating whether it is a tension or a thrust.

SOLUTION

Solution. Step 1 – Assumptions: car and trailer are particles; the tow-bar is light and rigid; the road is horizontal.

Step 2 – Diagrams: separate diagrams for car and trailer (see figure above).

Step 3 – Direction of motion: forwards, common acceleration a .

Step 4 – Newton's second law.

For the trailer (400 kg), the only horizontal forces are the tow-bar pull T and the resistance:

$$T - 200 = 400a. \quad (1)$$

For the car (1200 kg), the driving force $D = 2400$ N acts forwards, the resistance 600 N and the tow-bar reaction T act backwards:

$$2400 - 600 - T = 1200a. \quad (2)$$

Step 5 – Solve. Adding (1) and (2) gives the system equation:

$$2400 - 600 - 200 = 1600a \Rightarrow a = \frac{1600}{1600} = 1 \text{ m s}^{-2}.$$

From (1): $T = 200 + 400(1) = 600$ N.

Answers: (a) $a = 1 \text{ m s}^{-2}$; (b) the tow-bar is in tension of 600 N (it pulls the trailer forward and drags the car backward). Since $T > 0$, the bar is pulling, not pushing.

EXERCISE 1 A pickup and a taxi

A pickup of mass 1500 kg tows a broken-down taxi of mass 900 kg along a horizontal road with a light chain. The driving force is 3600 N; resistances are 750 N on the pickup and 450 N on the taxi. Find the acceleration and the tension in the chain. What happens to the tension if the pickup brakes gently?

SOLUTION Exercise 1

Newton's second law for the taxi: $T - 450 = 900a$. For the pickup: $3600 - 750 - T = 1500a$. Adding: $2400 = 2400a$, so $a = 1 \text{ m s}^{-2}$ and $T = 450 + 900 = 1350$ N. If the pickup brakes, the taxi pushes on the pickup; a chain cannot push, so it goes slack ($T = 0$) and the taxi

runs free until the chain tightens again.

DID YOU KNOW? Why tow-bars instead of ropes?

Road-safety rules in many countries require a rigid tow-bar rather than a rope for towing on fast roads: with a rope, the moment the towing vehicle slows down, the towed vehicle keeps going, the rope goes slack, and a collision follows. A rigid bar transmits thrust as well as tension, so the towing vehicle can help brake the towed one — exactly the string-versus-rod distinction of this section.

KEY POINTS Key points of Section 1

- Connected particles move with the same speed and same magnitude of acceleration when linked by a taut inextensible string.
- A light string over a smooth pulley has the same tension throughout; the pulley changes only the direction of the force.
- A string pulls only and may go slack ($T = 0$); a rod or tow-bar can also push (thrust).
- Always draw one free-body diagram per particle and write $F = ma$ for each, then solve simultaneously.
- Tension forces on the two bodies form a Newton third-law pair: equal magnitude, opposite direction.
- Check that $T \geq 0$ for a string; if $T < 0$ the string is slack and the particles move independently.

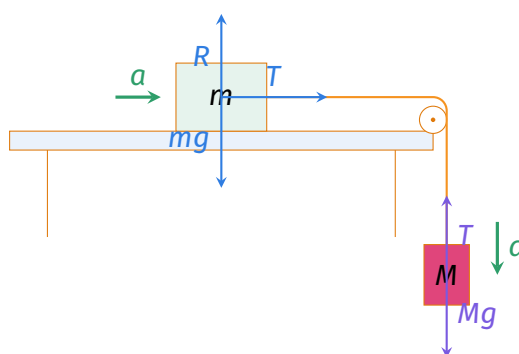
2 SECTION 2: MOTION ON SMOOTH AND ROUGH HORIZONTAL SURFACES

In Section 1 we learnt to model a system of connected particles: light inextensible strings, smooth pulleys, and the key fact that the **tension** is the same throughout a single string while both particles share the same **acceleration** magnitude. We now put these ideas to work in the most classic configuration of the GCE Advanced Level examination: one particle resting on a **horizontal table**, joined by a string passing over a pulley at the edge to a second particle hanging freely. We first treat the smooth table, then introduce friction, and finish with towing problems and the use of the constant-acceleration (suvat) formulae.

2.1 The Smooth Table with a Hanging Particle

EXPERIMENT Setting up the situation

Place a block of mass m on a polished table. Tie a light string to it, pass the string over a small pulley fixed at the edge of the table, and attach a mass M hanging freely. Release the system. The hanging mass falls, the block slides towards the edge, and both move together with the same acceleration because the string is inextensible. Our task is to predict that acceleration and the tension in the string.



Block of mass m on a smooth table, connected over a smooth pulley at the edge to a hanging mass M . Forces on each particle are shown.

Analysing each particle separately.

For the block on the table there is no vertical motion, so the vertical forces balance: $R = mg$. Horizontally the only force is the tension, so Newton's second law gives

$$T = ma. \quad (1)$$

For the hanging mass, taking the downward direction (the direction of its motion) as positive:

$$Mg - T = Ma. \quad (2)$$

Adding (1) and (2) eliminates T :

$$Mg = (M + m)a \quad \Rightarrow \quad a = \frac{Mg}{M + m}, \quad T = ma = \frac{Mmg}{M + m}.$$

METHOD Smooth table: direct formulae

For a hanging mass M driving a mass m on a smooth horizontal table:

1. Acceleration: $a = \frac{Mg}{M + m}$ — driving weight divided by total mass.

2. Tension: $T = \frac{Mmg}{M+m}$, obtained by substituting a into $T = ma$.

3. Always check: $T < Mg$ (the hanging mass is accelerating downwards, so its weight must exceed the tension).

This whole-system shortcut is valid whenever the string stays taut, the pulley is smooth, and no other horizontal force (such as friction) acts on the table particle.

WORKED EXAMPLE 2 Smooth table: finding a and T

A particle P of mass 2 kg rests on a smooth horizontal table. It is connected by a light inextensible string passing over a smooth pulley at the edge of the table to a particle Q of mass 3 kg hanging freely. The system is released from rest. Find the acceleration of the system and the tension in the string. Take $g = 9.8 \text{ m s}^{-2}$.

SOLUTION

Solution. For P (horizontal): $T = 2a$. For Q (downwards positive): $3g - T = 3a$. Adding:

$$3g = 5a \implies a = \frac{3 \times 9.8}{5} = 5.88 \text{ m s}^{-2}.$$

Then $T = 2a = 11.76 \text{ N}$. Check: $T = 11.76 < 3g = 29.4$ ✓, and $3g - T = 17.64 = 3 \times 5.88$ ✓.

WARNING The shortcut gives only a

A very common error is to write $a = \frac{Mg}{M+m}$ and then claim $T = Mg$ or $T = Ma$. The whole-system equation contains no tension at all — the internal forces cancel. You must return to the equation of one particle ($T = ma$ or $Mg - T = Ma$) to find T . Notice also that $T = \frac{Mmg}{M+m}$ is less than both Mg and (usually) what candidates guess.

2.2 Introducing Friction: the Rough Table

A real table — a wooden desk in a Yaoundé classroom, for instance — is not smooth. When the block slides (or tends to slide), the surface exerts a frictional force opposing the motion.

PROPERTY Normal reaction and the friction model

For a particle on a horizontal surface with no vertical acceleration, resolving vertically gives

$$R = mg.$$

The frictional force satisfies $F \leq \mu R$, where μ is the coefficient of friction. When the body is moving (or in limiting equilibrium, on the point of moving),

$$F = \mu R,$$

directed opposite to the motion (or to the tendency of motion). If the body is at rest and not on the point of moving, F takes whatever value below μR is needed to maintain equilibrium.

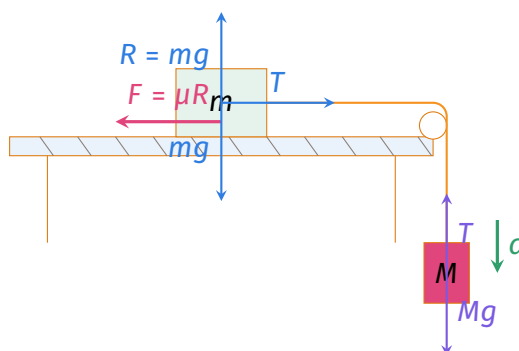
METHOD Test first: will the system move at all?

Before writing any equation of motion on a rough table:

1. Compute the maximum available friction $F_{\max} = \mu R = \mu mg$.
2. Compute the driving force, here the hanging weight Mg .
3. If $Mg \leq \mu mg$, the system does not move: friction adjusts to $F = Mg$ and the string tension is $T = Mg$ with $a = 0$.
4. If $Mg > \mu mg$, motion occurs; write the equations with $F = \mu R$ opposing the motion.

WARNING Do not write $F = \mu R$ blindly

Two mistakes are frequent. First, writing $F = \mu R$ for a smooth surface (where $F = 0$). Second, writing $F = \mu R$ when the body is at rest and the driving force is small: then $F < \mu R$, and assuming $F = \mu R$ can produce a negative acceleration — a contradiction telling you that your assumption of motion was wrong. The direction of F also matters: it always opposes the actual or impending motion, so it points away from the pulley here.



Rough table: friction $F = \mu R$ opposes the motion of the block towards the pulley, while vertically $R = mg$.

Equations of motion on the rough table.

Assuming motion occurs (block moving towards the pulley):

$$\text{Block: } T - F = ma, \quad F = \mu R = \mu mg; \quad \text{Hanging mass: } Mg - T = Ma.$$

Adding eliminates T :

$$Mg - \mu mg = (M + m)a \implies a = \frac{(M - \mu m)g}{M + m}, \quad T = M(g - a) = \frac{Mm(1 + \mu)g}{M + m}.$$

WORKED EXAMPLE 3 Rough table: full analysis

A block of mass 4 kg lies on a rough horizontal table with coefficient of friction $\mu = 0.25$. It is joined by a light inextensible string over a smooth edge pulley to a hanging mass of 6 kg. The system is released from rest. Take $g = 9.8 \text{ m s}^{-2}$.

1. Show that motion occurs.
2. Find the acceleration and the tension.
3. Find the speed of the hanging mass after it has fallen 0.8 m.

SOLUTION

Solution. (1) Maximum friction: $F_{\max} = \mu mg = 0.25 \times 4 \times 9.8 = 9.8 \text{ N}$. Driving weight: $Mg = 6 \times 9.8 = 58.8 \text{ N}$. Since $58.8 > 9.8$, motion occurs. (2) Block: $T - 9.8 = 4a$. Hanging mass: $58.8 - T = 6a$. Adding: $49 = 10a$, so $a = 4.9 \text{ m s}^{-2} > 0$ — consistent with our assumption of motion ✓. Then $T = 9.8 + 4(4.9) = 29.4 \text{ N}$. (3) Using $v^2 = u^2 + 2as$ with $u = 0$, $a = 4.9$, $s = 0.8$:

$$v^2 = 2 \times 4.9 \times 0.8 = 7.84 \implies v = 2.8 \text{ m s}^{-1}.$$

WORKED EXAMPLE 4 When the system does not move

The block of the previous example is replaced by one of mass 10 kg, with $\mu = 0.7$, the hanging mass remaining 6 kg. Describe the motion.

SOLUTION

Solution. $F_{\max} = \mu mg = 0.7 \times 10 \times 9.8 = 68.6$ N, while the driving weight is only $Mg = 58.8$ N. Since $58.8 < 68.6$, friction can balance the pull: the system remains at rest. The actual friction is $F = 58.8$ N $< \mu R$, and the tension is $T = Mg = 58.8$ N (the hanging mass is in equilibrium). Writing $a = \frac{(M - \mu m)g}{M + m}$ would give a negative value — the signal that the motion assumption fails.

2.3 Using the suvat Equations

Once the (constant) acceleration is known, all the constant-acceleration formulae apply to each particle, since both share a :

$$v = u + at, \quad s = ut + \frac{1}{2}at^2, \quad v^2 = u^2 + 2as, \quad s = \frac{1}{2}(u + v)t.$$

Typical examination questions ask for the time taken by the table particle to reach the pulley, or the speed of the hanging mass when it hits the floor. Remember that the two particles describe the same distance along the string while it remains taut.

WORKED EXAMPLE 5 Time to reach the edge

In the first worked example (smooth table, $m = 2$, $M = 3$, $a = 5.88 \text{ m s}^{-2}$), the block starts 1.5 m from the pulley. Find the time taken to reach the pulley and its speed on arrival.

SOLUTION

Solution. From $s = ut + \frac{1}{2}at^2$ with $u = 0$:

$$1.5 = \frac{1}{2}(5.88)t^2 \Rightarrow t^2 = \frac{3}{5.88} = 0.5102 \Rightarrow t \approx 0.714 \text{ s.}$$

Speed: $v = at = 5.88 \times 0.714 \approx 4.2 \text{ m s}^{-1}$ (check with $v^2 = 2as = 17.64$, $v = 4.2$ ✓).

2.4 Extension: Towing Problems with Resistance

The same logic governs a **car towing a trailer**, with the tow-bar playing the role of the string. The difference is that a tow-bar can push as well as pull: when the car brakes, the bar is in **compression** and exerts a **thrust**. Resistance forces (air resistance, rolling resistance) act on each vehicle, constant or proportional to speed; at GCE level they are usually given as constants.

METHOD Car and trailer

1. Draw separate force diagrams for the car and the trailer; mark driving force, resistances, and the force T in the tow-bar. Assume T is a tension (pull) — i.e. the bar pulls the trailer forward and the car backward.
2. Write $F = ma$ for each vehicle along the direction of motion (take the direction of motion as positive).
3. Add the equations to get a for the whole system, then back-substitute into one equation to find T .
4. If T comes out negative, the bar is in thrust (compression) of magnitude $|T|$ — typical under braking.

WORKED EXAMPLE 6 Braking car and trailer: thrust in the tow-bar

A car of mass 1200 kg tows a trailer of mass 400 kg along a horizontal road. The car brakes, producing a braking force of 6000 N; resistances of 200 N on the car and 100 N on the trailer also act. Find the deceleration and the force in the tow-bar, stating whether it is a tension or a thrust.

SOLUTION

Solution. Take the direction of motion as positive. All resistive forces are negative. Whole system:

$$-(6000 + 200 + 100) = (1200 + 400)a \Rightarrow -6300 = 1600a \Rightarrow a = -\frac{6300}{1600} = -3.9375 \text{ m s}^{-2}.$$

The deceleration is 3.9375 m s^{-2} . Now consider the trailer alone. Assume the tow-bar is in tension, so it pulls the trailer forward with force T (positive). The resistance on the trailer is 100 N backward.

$$T - 100 = 400 \times (-3.9375) = -1575 \Rightarrow T = -1475 \text{ N}.$$

Since T is negative, our assumption of tension is wrong; the bar is in compression (thrust) of magnitude 1475 N. The bar pushes the trailer backward (and the car forward) to slow the trailer down.

WARNING Sign conventions in towing

Decide at the start which way is positive and assume the tow-bar force is a tension (pull). Keep that convention for both vehicles: the bar exerts equal and opposite forces on car

and trailer (Newton's third law). A negative result for T means the bar is in thrust. Mixing conventions is the most common source of sign errors in braking questions.

DID YOU KNOW? Friction in daily life

The friction model $F = \mu R$ is an approximation (Amontons–Coulomb law). Real friction depends on contact area, temperature, and speed. In Cameroon, think of a taxi braking on a wet road in Douala: μ drops dramatically, increasing stopping distance. Engineers design brake systems to maximise μ while avoiding wheel lock (which switches from static to kinetic friction, reducing μ). This is why anti-lock braking systems (ABS) are vital for safety.

KEY POINTS Key points of Section 2

- Smooth table with hanging mass M driving mass m : $a = \frac{Mg}{M+m}$, $T = \frac{Mmg}{M+m}$.
- Rough table: first test Mg against μmg . Motion occurs only if $Mg > \mu mg$; then $a = \frac{(M - \mu m)g}{M+m}$ and $F = \mu R$ opposes the motion.
- On a horizontal surface with no vertical acceleration, $R = mg$.
- The whole-system equation yields a only; tension requires a separate single-particle equation.
- With constant a found, use the suvat formulae for times, speeds and distances.
- In towing problems, assume tension in the tow-bar; a negative result signals a thrust (compression), typical under braking.

EXERCISE 2 Practice

1. A mass of 5 kg on a smooth table is connected over a smooth edge pulley to a hanging mass of 3 kg. Find the acceleration, the tension, and the speed of the system after 0.5 s from rest. ($g = 9.8 \text{ m s}^{-2}$)
2. A block of mass 8 kg rests on a rough horizontal table, $\mu = 0.3$, connected to a hanging mass of 2 kg. Determine whether the system moves; if it does, find a and T ; if not, find the friction and the tension.
3. Repeat question 2 with a hanging mass of 5 kg, and find how long the block takes to travel 1.2 m from rest.

4. A car of mass 1000 kg towing a trailer of mass 250 kg accelerates with driving force 3200 N against total resistances of 450 N on the car and 150 N on the trailer. Find the acceleration and the tension in the tow-bar.

SOLUTION Exercise 1

$$a = \frac{3g}{8} = 3.675 \text{ m s}^{-2}; T = 5a = 18.375 \text{ N}; v = at = 3.675 \times 0.5 \approx 1.84 \text{ m s}^{-1}.$$

SOLUTION Exercise 2

$$F_{\max} = 0.3 \times 8 \times 9.8 = 23.52 \text{ N}; \text{ driving weight} = 2g = 19.6 \text{ N} < 23.52 \text{ N. No motion: } a = 0, \\ F = 19.6 \text{ N}, T = 19.6 \text{ N}.$$

SOLUTION Exercise 3

$$5g = 49 > 23.52, \text{ so motion occurs. } a = \frac{(5 - 0.3 \times 8) \times 9.8}{13} = \frac{2.6 \times 9.8}{13} = 1.96 \text{ m s}^{-2}; T = \\ 5(g - a) = 5 \times 7.84 = 39.2 \text{ N. From } s = \frac{1}{2}at^2: t = \sqrt{\frac{2 \times 1.2}{1.96}} = \sqrt{1.2245} \approx 1.11 \text{ s}.$$

SOLUTION Exercise 4

$$\text{Whole system: } 3200 - 450 - 150 = 1250a, \text{ so } a = \frac{2600}{1250} = 2.08 \text{ m s}^{-2}. \text{ Trailer: } T - 150 = \\ 250 \times 2.08 = 520, \text{ hence } T = 670 \text{ N (tension, as expected under acceleration).}$$

3 SECTION 3: MOTION ON INCLINED PLANES

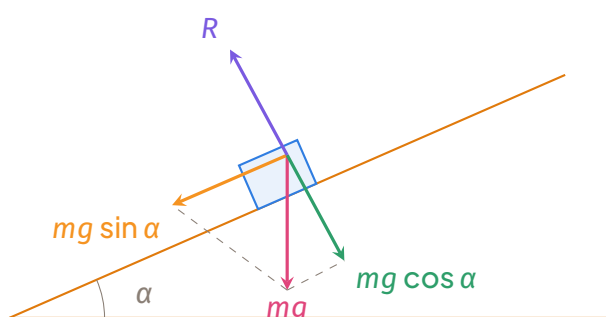
In Section 2 we studied connected particles on horizontal surfaces, where the weight of a block was completely balanced by the normal reaction ($R = mg$) and only the tension and friction acted along the direction of motion. Life in Cameroon, however, is rarely horizontal: think of a truck climbing the steep road to Dschang, a wheelbarrow pushed up a plank onto a lorry in a Douala warehouse, or a crate sliding down a ramp. On a slope, the weight itself has a component along the surface, and this changes everything: the normal reaction, the friction, and the driving force of the motion. This section teaches you to handle such problems with complete confidence.

3.1 Resolving the Weight on an Inclined Plane

Observation. Place a book on a flat table and tilt the table slowly. At first nothing happens; then, beyond a certain angle, the book slides. Gravity has not changed — what has changed is the direction in which gravity can push the book relative to the surface. To understand the motion we must split the weight into two perpendicular parts: one pulling the object along the plane, and one pressing it into the plane.

Consider a particle of mass m on a plane inclined at an angle α to the horizontal. The weight mg acts vertically downwards. We resolve it into:

- a component **parallel to the plane**, pointing down the slope;
- a component **perpendicular to the plane**, pressing into the surface.



Resolving the weight on a plane inclined at α : the component $mg \sin \alpha$ acts down the plane, $mg \cos \alpha$ presses into the plane, and the normal reaction R acts perpendicular to the surface.

The geometry is the key point. The angle between the vertical (direction of mg) and the normal to the plane is exactly α , because the normal is tilted by α from the vertical just as the plane is tilted by α from the horizontal. Hence, in the right-angled triangle formed by the weight and its two components:

$$\text{component down the plane} = mg \sin \alpha, \quad \text{component into the plane} = mg \cos \alpha.$$

PROPERTY Weight components and normal reaction on an inclined plane

For a particle of mass m resting on a plane inclined at angle α to the horizontal:

- the component of weight down the plane is $mg \sin \alpha$;
- the component of weight perpendicular into the plane is $mg \cos \alpha$;
- since there is no acceleration perpendicular to the plane (the particle neither sinks into nor lifts off the surface), resolving perpendicular to the plane gives

$$R = mg \cos \alpha.$$

This result is examinable and must be justified by resolving perpendicular to the plane.